A LOCAL PROPERTY OF RING MAPS HOLDS ON EVERY PAIR OF AFFINE OPENS WITH THE IMAGE OF ONE INSIDE THE OTHER

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

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Editorial note. Statement span 2453-2462 and proof span 2464-2492 come from the halfA pool card operations/pool-cards/stacks-morphisms-halfA.json and were re-read in the pinned chapter file before this card was written.

Editorial note. The proof refers to the three conditions of the cited definition by 'Definition (1)(a)', '(1)(b)' and '(1)(c)'; those references are the author's and are kept verbatim.

Editorial note. 2 cross-references occur inside the counted spans; they are listed in external_references as navigation pointers only.

Allowed standard prerequisites (editorial).

- The definition of a local property of ring maps with its conditions (a) on standard localizations of the target, (b) on standard localizations of the source and (c) on finite covers of the source. Cited repeatedly at proof lines 2687, 2696, 2699 and 2704. Every shrinking step in the proof is an application of one of these three conditions.
- Two affine opens of a scheme with a common point have a common neighbourhood that is standard affine open in both, cited from the schemes chapter. Cited at proof lines 2684-2686 and 2691-2693. Produces the standard affine neighbourhoods used to invoke conditions (a) and (b) of locality.
- An affine scheme is quasi-compact, so a cover of it by open pieces has a finite subcover. Used at proof lines 2700-2703. Turns the pointwise reductions into a finite cover to which condition (c) of locality can be applied.

Source reference. Original definition 01SR: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2348–2348. Same-chapter reference used in the retained author text at source line(s) 2471, 2480, 2483, 2490, 2699, 3867, 3875, 4122, 5812, 7074, 7695, 8116. It is supplied only as a navigation pointer so

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Source reference. Original lemma 01IW: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 1993–1993. Same-chapter reference used in the retained author text at source line(s) 2470, 2476, 14942. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Lemma 0.1. *Let $f : X \to S$ be a morphism of schemes. Let P be a property of ring maps. Let U be an affine open of X, and V an affine open of S such that $f(U) \subset V$. If f is locally of type P and P is local, then $P(\mathcal{O}_S(V) \to \mathcal{O}_X(U))$ holds.*

Proof. As f is locally of type P for every $u \in U$ there exists an affine open $U_u \subset X$ mapping into an affine open $V_u \subset S$ such that $P(\mathcal{O}_S(V_u) \to \mathcal{O}_X(U_u))$ holds. Choose an open neighbourhood $U'_u \subset U \cap U_u$ of u which is standard affine open in both U and U_u, see Schemes, Lemma 01IW. By Definition 01SR (1)(b) we see that $P(\mathcal{O}_S(V_u) \to \mathcal{O}_X(U'_u))$ holds. Hence we may assume that $U_u \subset U$ is a standard affine open. Choose an open neighbourhood $V'_u \subset V \cap V_u$ of $f(u)$ which is standard affine open in both V and V_u, see Schemes, Lemma 01IW. Then $U'_u = f^{-1}(V'_u) \cap U_u$ is a standard affine open of U_u (hence of U) and we have $P(\mathcal{O}_S(V'_u) \to \mathcal{O}_X(U'_u))$ by Definition 01SR (1)(a). Hence we may assume both $U_u \subset U$ and $V_u \subset V$ are standard affine open. Applying Definition 01SR (1)(b) one more time we conclude that $P(\mathcal{O}_S(V) \to \mathcal{O}_X(U_u))$ holds. Because U is quasi-compact we may choose a finite number of points $u_1, \ldots, u_n \in U$ such that

$$U = U_{u_1} \cup \ldots \cup U_{u_n}.$$

By Definition 01SR (1)(c) we conclude that $P(\mathcal{O}_S(V) \to \mathcal{O}_X(U))$ holds. □